Project Plan Document

Project: Skill/Interest-Based Secure Matching Platform for Erasmus and Local Students

Course: BIL 482 Design Patterns – TOBB University of Economics and Technology, Fall 2026

Instructor: Dr. Çiğdem Avcı

Assignment: Project Assignment 1 (PA1)

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1. Project Scope

This project will deliver a web-based matching platform connecting Erasmus and local students based on shared interests, hobbies and language goals.

- **Included:** user registration, interest/skill-based profiling, configurable matching, event creation and participation, user notifications, and a multi-level inappropriate content reporting system. The user interface will be developed in Turkish, English and Spanish. eduroam authentication and translation APIs will be simulated using mock implementations via Adapters.
- **Not included:** native iOS or Android applications, deployment into a real production environment with live eduroam or commercial translation API billing, real-time chat, payment gateways, and formal legal GDPR/KVKK auditing.

2. Project Organization

Member	Role	Responsibilities
Alberto Alvarez Polo (261118082)	Architecture lead	Project Definition Document; Software Architecture Overview; implementation of the Presentation Layer and the eduroam/translation Adapters.
Alejandro Blas Piris (261118081)	Planning & patterns lead	Project Plan Document; Pattern–Use Case Mapping Matrix; implementation of the Strategy (UC2) and Chain of Responsibility (UC4) parts of the Business Layer.
Carlos Molina Perez (261118083)	Requirements & integration lead	Software Requirements Specification and integration of the architecture and pattern sections; implementation of UC1 (Factory Method) and UC3 (Observer), Data Access Layer and acceptance tests.

All members take part in code reviews, testing and the final demonstration. The instructor, Dr. Çiğdem Avcı, is the main stakeholder and evaluates every deliverable.

3. Objectives

The management and execution objectives for this project are to:

- Establish a structured, predictable timeline ensuring all Project Assignments (PA1, PA2, PA3) and the final source code are delivered on or before the deadlines.
- Ensure clear tracking of individual contributions via GitHub for grading transparency.
- Maintain an organized layered architecture to allow concurrent development by all three team members without codebase conflicts.
- Keep the documentation, the code and the tests traceable through the use case numbering (UC1–UC4) defined in the SRS.

4. Key Phases and Timeline

The project runs during the Fall 2026 semester. The dates below are the team's planned dates; if the instructor publishes different deadlines, the plan is updated through the Change Management Plan (Section 8) and the internal deadlines are moved accordingly.

4.1 Phases

Phase	Description	Start	End
P1 – Requirements & Initial Design (PA1)	Gather requirements, define the architecture, select patterns; Project Definition, Project Plan and SRS.	28 Sep 2026	26 Oct 2026
P2 – Detailed Design (PA2)	In-depth application of patterns to one selected use case; preparation of the Design Pattern Workshop presentation.	27 Oct 2026	16 Nov 2026
P3 – Final Architecture & Technologies (PA3)	Finalization of the project design, technologies and integration strategy.	17 Nov 2026	07 Dec 2026
P4a – Implementation	Coding of UC1–UC4, the Layered Architecture and the five design patterns. Starts with the project skeleton while PA2 and PA3 are being prepared.	02 Nov 2026	20 Dec 2026
P4b – Testing	Execution of the acceptance tests (AT-UC1-01 … AT-UC4-03), bug fixing and usability test with students.	07 Dec 2026	27 Dec 2026
P5 – Closure	Final source code push to GitHub, live presentation and working demonstration, project evaluation.	28 Dec 2026	08 Jan 2027

4.2 Gantt Chart

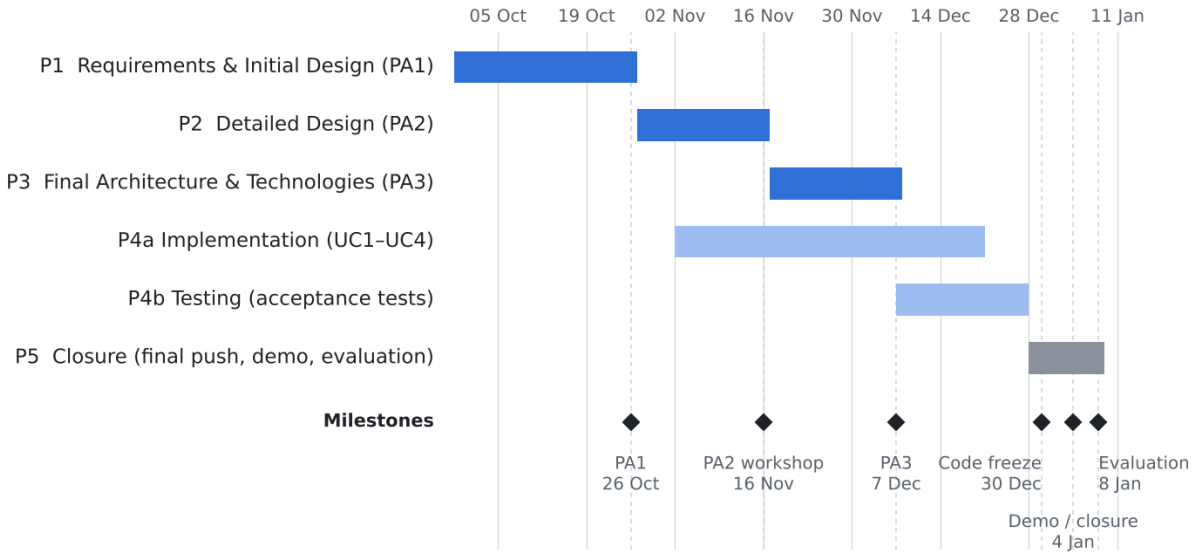


Figure 1 – Gantt chart of the project phases and milestones

4.3 Deliverables and Delivery Dates

Milestone	Date	Deliverables
PA1 delivery	26 Oct 2026	Project Definition, Project Plan, Software Requirements Specification (with Software Architecture Overview and Pattern–Use Case Mapping) on GitHub
PA2 – Design Pattern Workshop	16 Nov 2026	One selected use case with detailed pattern application (class and sequence diagrams) and its presentation
PA3 delivery	07 Dec 2026	Final project design and technologies document
Code freeze	30 Dec 2026	Final source code of UC1–UC4 pushed to GitHub, test results
Closure date – final presentation & demo	04 Jan 2027	Working demo of the platform and final documentation
Project evaluation date	08 Jan 2027	Team retrospective: success criteria and acceptance criteria checked, lessons learned recorded in the repository

Internal deadlines are set three days before each official date, so that the integrated deliverable can be reviewed by all members before submission.

5. Resource Planning

Resource	Use	How it is accessed
Web stack (HTML/CSS/JavaScript)	Presentation Layer	Free and open source; installed locally
Object-oriented backend language	Business Layer and design patterns	Free SDK and IDE installed on each member's computer; final choice confirmed in PA3
Relational database	Data Access Layer	Free/open-source database run locally for development and in the demo environment
Mock eduroam and translation services	External services behind Adapters	Implemented by the team inside the project; no external account needed
Git and GitHub	Version control, issue tracking, submissions	Shared repository; each member commits from an individual GitHub account; the instructor has access
Google Drive	Drafts and document aggregation	Shared team folder
Draw/diagram tools	UML and architecture diagrams	Free online or open-source tools

6. Risk Management

Risk	Probability / impact	Mitigation
Schedule delays	Medium / High	Internal deadlines three days before each official deadline; weekly progress check.
Technical difficulties	Medium / Medium	Development is isolated into independent services (UserService, EventService, etc.), so issues in one area do not block the others.
Team member absence (illness)	Low / High	Code and documents are centralized on GitHub and Google Drive so any member can continue the work.
Scope creep	Medium / Medium	Any new feature goes through the Change Management Plan; features out of scope (chat, mobile apps) are rejected.
Merge conflicts	Medium / Low	Each member works on a separate layer/service and integrates through small, frequent commits.

7. Communication Plan

Within the team. The team maintains continuous daily communication through a dedicated WhatsApp group for immediate query resolution. A formal weekly synchronization meeting is held to review GitHub progress, align on architecture decisions and plan upcoming tasks; decisions are recorded as GitHub issues.

With stakeholders. The instructor is contacted through the course's official channels and by e-mail when a requirement is unclear. Progress and deliverables are communicated through the GitHub repository, which the instructor can access at any time, and through the Design Pattern Workshop (PA2) and the final presentation and demo.

8. Change Management Plan

Any proposed change to the project scope, core use cases, assigned design patterns or schedule must be raised during the weekly synchronization meeting (or in the WhatsApp group if urgent) and recorded as a GitHub issue. A change is only adopted if all three team members agree it is feasible. Accepted changes must be reflected immediately in the SRS and, if they affect dates, in this plan.

9. Budget Plan

The project has no monetary budget: all tools are free. The budget is therefore the team's effort, estimated at about 60 hours per member (180 hours in total), distributed to balance planning and execution:

Area	Share	Effort (h)	Phases	Allocation
Design & Documentation	30%	54	P1, P2, P3	PA1, PA2 and PA3 deliverables; each member leads one document per assignment (≈18 h each)
Implementation	50%	90	P4a	Code for UC1–UC4 and the patterns; ≈30 h per member following the responsibilities in Section 2

Area	Share	Effort (h)	Phases	Allocation
Testing & Closure	20%	36	P4b, P5	End-to-end testing, bug fixing and preparation of the final demonstration; ≈12 h per member
Total	**100%**	**180**		

10. Acceptance Tests and Acceptance Criteria

The project will be considered successful if it meets the following criteria before the final demonstration. The acceptance tests that verify them are defined per use case in the SRS (Section 8, AT-UC1-01 … AT-UC4-03).

Criterion	Description	Verified by
Functional end-to-end	All four core use cases (UC1–UC4) execute without critical errors in a live demo.	AT-UC1-01 … AT-UC4-03
Traceable design	Every implemented use case explicitly uses at least one documented design pattern, clearly identifiable in the source code.	Code review against SRS Section 7
Usability benchmark	A user can complete registration and receive their first match in under 5 minutes.	Timed test with students (NFR-U.1)
Multilingual support	The interface switches correctly between Turkish, English and Spanish.	Manual test of all screens (NFR-U.3)

11. Task Matrix

Task	Responsible
Project scope, organization and objectives	Alejandro Blas Piris
Key phases and timeline	Alejandro Blas Piris
Resources, risk and communication plans	Alejandro Blas Piris
Change management, budget and acceptance tests	Alejandro Blas Piris
Timeline dates, Gantt chart, deliverable/closure/evaluation dates	Carlos Molina Perez
Resource access, stakeholder communication and effort allocation	Carlos Molina Perez
Review of the plan against the Project Definition	Alberto Alvarez Polo
Final revision and formatting	Carlos Molina Perez